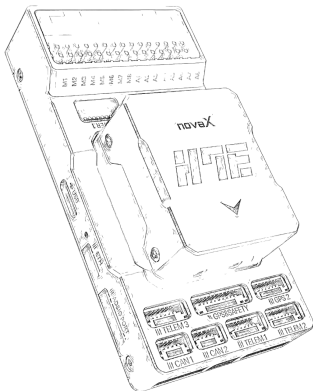


AF-H7E

User Guide



AF-H7E Overview



Modular STM32H753 flight controller on the Pixhawk FMUv6x architecture with three IMUs, an RM3100 compass, two ICP-20100 barometers, 8 FMU + 8 IOMCU outputs, 2 CAN ports and Ethernet.

AF-H7E Specifications

• Basic Specifications

MCU	STM32H753, ARM Cortex-M7, 480 MHz
RAM / Flash	1 MB / 2 MB
IMU	BMI088 (Bosch, accel / gyro)
Secondary IMU	ICM-42688-P, ICM-20649 (TDK InvenSense)
IMU Heater	Built-in heater on the IMU board, held at 45 °C (BRD_HEAT_TARG)
Magnetometer	RM3100 (PNI)
Barometer	2× ICP-20100 (TDK InvenSense)
Operating Voltage	4.75 – 5.7 V (Rated 5 V)
USB Input	4.75 – 5.25 V
Servo Rail	0 – 9.9 V

AF-H7E Specifications

• Basic Specifications

PWM Output	8 FMU + 8 IOMCU channels
RC Input	S.Bus, PPM, DSM / Spektrum
RSSI Input	Analog / PWM
UART	7 peripheral serial interfaces + dedicated IOMCU link
Serial Mapping	SERIAL1 = UART7 · SERIAL2 = UART5 · SERIAL3 = USART1 · SERIAL4 = UART8 · SERIAL5 = USART2 · SERIAL6 = UART4 · SERIAL7 = USART3 (USB = SERIAL0)
I ² C	4 configured buses including internal sensors; 3 exposed on multifunction ports
CAN	2 Port

AF-H7E Specifications

• Basic Specifications

ADC	VBat/Current + Aux Analog Input
Size	45 × 90 × 29.2 mm
Mount Hole	Pixhawk FMUv6x Standard
Weight	99 g (Core 43g + Baseboard 56g)
Operating Temp	-20 ~ +85 °C
Supported F/W	novaX ArduPilot (Copter and Plane releases)
Ethernet	100 Mbps x 1 Port

AF-H7E Pinout

• Pinout

AF-HZE Circuit

Each table lists the gene info → right margin as you see them in the picture & points to...

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Top view — map (PMM needed at the top)

March 2018 — 1 • 1st Semester

[illegible][illegible]

Left side — the front of the sound is to the right

1004-2

Figure 1 shows a schematic diagram of the experimental setup. It includes a top-down view of a grey rectangular block with two blue rectangular components (labeled 'S1' and 'S2') and a black circular component (labeled 'S3') on its top surface. Below the block, two tables are shown. The first table is labeled 'S1' and the second is labeled 'S2'. Both tables have columns for 'X', 'Y', and 'Z' coordinates.

Right side — the rear (PWH header) is in the right

Friend and

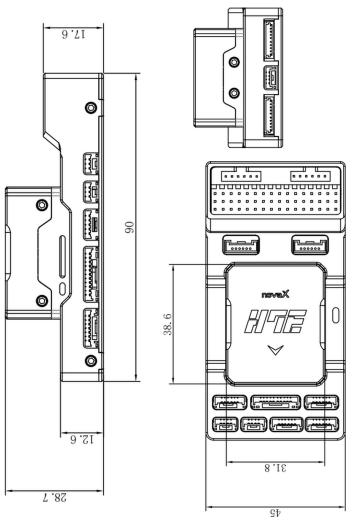
Figure 1 shows the test setup. The top table, labeled 'VISA Device', has columns for 'Device', 'Port', 'Type', 'Manufacturer', 'Model', 'Serial', and 'Status'. The bottom section contains two 'VISA' tables, each with columns for 'Device', 'Port', 'Type', 'Manufacturer', 'Model', 'Serial', and 'Status'. Blue dashed lines indicate connections from the 'VISA Device' table to the top module and from the first 'VISA' table to the bottom module.

AF-H7E Pinout

• Dimensions

AF-H7E Dimensions

Millimetres. Side view (top), top view and front view (bottom).



AF-H7E Pinout

• Pinout Notes

I2C bus numbering on the multi-function ports: the GPS & Safety port carries I2C1, the GPS2 port carries I2C2, and the UART4 port carries I2C3.

AF-H7E Pin Definition

• POWER 1

Molex Micro-Lock Plus 6P (1.25 mm) · top, left of the FMU ·
Primary power input · I2C1 power monitor

Pin	Signal	Function
1	VCC_IN	5 V supply input from the power module
2	VCC_IN	5 V supply input from the power module
3	SCL	Power-module I2C clock (voltage / current monitor)
4	SDA	Power-module I2C data (voltage / current monitor)
5	GND	Ground
6	GND	Ground

AF-H7E Pin Definition

• POWER 2

Molex Micro-Lock Plus 6P (1.25 mm) · top, right of the FMU ·
Redundant power input · I2C2 power monitor

Pin	Signal	Function
1	VCC_IN	5 V supply input from the second power module
2	VCC_IN	5 V supply input from the second power module
3	SCL	Power-module I2C clock (voltage / current monitor)
4	SDA	Power-module I2C data (voltage / current monitor)
5	GND	Ground
6	GND	Ground

AF-H7E Pin Definition

• POWER C1

6P · top, rear left (on the PWM header block) · CAN power module input · CAN1

Pin	Signal	Function
1	GND	Ground
2	GND	Ground
3	CAN_L	CAN1 bus low (same bus as the CAN 1 port)
4	CAN_H	CAN1 bus high
5	VCC_IN	5 V supply input from the CAN power module
6	VCC_IN	5 V supply input from the CAN power module

AF-H7E Pin Definition

• POWER C2

6P · top, rear right (on the PWM header block) · CAN power module input · CAN2

Pin	Signal	Function
1	GND	Ground
2	GND	Ground
3	CAN_L	CAN2 bus low (same bus as the CAN 2 port)
4	CAN_H	CAN2 bus high
5	VCC_IN	5 V supply input from the second CAN power module
6	VCC_IN	5 V supply input from the second CAN power module

AF-H7E Pin Definition

• TELEM 1

JST-GH 6P · front, lower row · SERIAL1 · UART7 · default
MAVLink2

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	CTS	Clear to send — hardware flow control input
5	RTS	Request to send — hardware flow control output
6	GND	Ground

AF-H7E Pin Definition

• TELEM 2

JST-GH 6P · front, lower row · SERIAL2 · UART5 · default
MAVLink2

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	CTS	Clear to send — hardware flow control input
5	RTS	Request to send — hardware flow control output
6	GND	Ground

AF-H7E Pin Definition

• TELEM 3

JST-GH 6P · front, upper row · SERIAL5 · USART2

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	CTS	Clear to send — hardware flow control input
5	RTS	Request to send — hardware flow control output
6	GND	Ground

AF-H7E Pin Definition

• GPS & SAFETY

JST-GH 10P · front, upper row · SERIAL3 · USART1 + I2C1 · default GPS

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → GPS)
3	RX	UART receive (GPS → FC)
4	SCL	I2C1 clock — compass in the GPS module
5	SDA	I2C1 data
6	SAFETY_SW	Safety switch input
7	SAFETY_LED	Safety switch LED output
8	VCC_3V3	3.3 V supply for the safety switch
9	BUZZER	Buzzer output
10	GND	Ground

AF-H7E Pin Definition

• GPS 2

JST-GH 6P · front, upper row · SERIAL4 · UART8 + I2C2 · default GPS

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → GPS)
3	RX	UART receive (GPS → FC)
4	SCL	I2C2 clock — compass in the GPS module
5	SDA	I2C2 data
6	GND	Ground

AF-H7E Pin Definition

• UART 4

JST-GH 7P · right side, front · SERIAL6 · UART4 + I2C3

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	SCL	I2C3 clock
5	SDA	I2C3 data
6	NFC_GPIO	General-purpose I/O
7	GND	Ground

• CAN 1

JST-GH 4P · front, lower row · CAN1 · DroneCAN

Pin	Signal	Function
1	VCC	5 V output to CAN peripherals (not a power input)
2	CAN_H	CAN bus high
3	CAN_L	CAN bus low
4	GND	Ground

AF-H7E Pin Definition

• CAN 2

JST-GH 4P · front, lower row · CAN2 · DroneCAN

Pin	Signal	Function
1	VCC	5 V output to CAN peripherals (not a power input)
2	CAN_H	CAN bus high
3	CAN_L	CAN bus low
4	GND	Ground

• DSM / SBUS RC

JST-GH 5P · right side · RC input through the IOMCU · SBUS / DSM

Pin	Signal	Function
1	VCC	5 V receiver supply
2	RC_IN	SBUS / DSM receiver input
3	RSSI	Analog RSSI input
4	VCC_3V3	3.3 V supply for DSM / Spektrum satellite receivers
5	GND	Ground

AF-H7E Pin Definition

• PPM IN

JST-GH 3P · right side · PPM receiver input

Pin	Signal	Function
1	VCC	5 V receiver supply
2	PPM	PPM receiver input
3	GND	Ground

• SBUS OUT

JST-GH 3P · right side, rear · SBUS output from the IOMCU

Pin	Signal	Function
1	NC	Not connected
2	SBUS_OUT	SBUS servo output
3	GND	Ground

AF-H7E Pin Definition

• PWM OUT M1–M8

3 × 16 pin header · rows about 2.54 mm (servo plug S · + · -), channels 2.6 mm apart on the base PCB · columns 1–8 (left) · MAIN outputs from the IOMCU · SERVO1–8

Pin	Signal	Function
S	M1 ... M8	Motor / servo output signal, one per channel (rear row)
+	V_SERVO	Servo rail 0 – 9.9 V, bussed across all channels — supplied externally (e.g. BEC)
-	GND	Ground (front row)

• PWM OUT A1–A8

3 × 16 pin header · rows about 2.54 mm (servo plug S · + · -), channels 2.6 mm apart on the base PCB · columns 9–16 (right) · AUX outputs from the FMU · SERVO9–16

Pin	Signal	Function
S	A1 ... A8	Motor / servo output signal, one per channel (rear row)
+	V_SERVO	Servo rail 0 – 9.9 V, bussed across all channels — supplied externally (e.g. BEC)
-	GND	Ground (front row)

AF-H7E Pin Definition

• ETHERNET

JST-GH 4P · left side · 100BASE-T

Pin	Signal	Function
1	RX-	Ethernet receive pair -
2	RX+	Ethernet receive pair +
3	TX-	Ethernet transmit pair -
4	TX+	Ethernet transmit pair +

• USB-C

USB Type-C · left side, rear · SERIAL0 · USB 2.0 Full Speed

Pin	Signal	Function
A4 B4 A9 B9	VBUS	5 V from USB — configuration and firmware update on the bench
A6 B6	D+	USB data +
A7 B7	D-	USB data -
A5 B5	CC1 / CC2	Configuration channel (device role)
A1 B1 A12 B12	GND	Ground

AF-H7E Pin Definition

• USB

JST-GH 4P · front end, middle · USB 2.0 — same data lines as the USB-C port (use one at a time)

Pin	Signal	Function
1	VBUS	5 V USB supply
2	D-	USB data -
3	D+	USB data +
4	GND	Ground

• AD & IO

JST-GH 8P · left side, front · FMU auxiliary inputs and outputs

Pin	Signal	Function
1	VCC	5 V output
2	CAP1	FMU timer capture input
3	BOOTLOAD ER	FMU bootloader request input
4	RST_REQ	FMU reset request input
5	nARMED	Armed status output (low = armed)
6	ADC_3V3	Analog input, 0 – 3.3 V
7	ADC_6V6	Analog input, 0 – 6.6 V (divided)
8	GND	Ground

AF-H7E Pin Definition

• SPI 6

JST-GH 11P · right side · External SPI6 — not enabled in novaX firmware

Pin	Signal	Function
1	VCC	5 V output
2	SCK	SPI clock
3	MISO	SPI data in
4	MOSI	SPI data out
5	CS1	Chip select 1
6	CS2	Chip select 2
7	SYNC	Sync output
8	DRDY1	Data-ready input 1
9	DRDY2	Data-ready input 2
10	nRESET	Reset output
11	GND	Ground

AF-H7E Pin Definition

• FMU DEBUG

JST-SH 10P · front end, left · Pixhawk Debug Full · SERIAL7 ·
USART3 · console + SWD

Pin	Signal	Function
1	VCC_3V3	3.3 V reference for the debug probe
2	CONSOLE_TX	Debug console transmit (FC → probe)
3	CONSOLE_RX	Debug console receive (probe → FC)
4	SWDIO	SWD data — bootloader programming and firmware recovery
5	SWCLK	SWD clock
6	SWO	Trace output (shared with SPI6 SCK)
7	NFC_GPIO	General-purpose I/O
8	PH11	General-purpose I/O
9	nRST	FMU reset
10	GND	Ground

AF-H7E Pin Definition

• IO DEBUG

JST-SH 10P · front end, right · IOMCU console + SWD

Pin	Signal	Function
1	VCC_3V3	3.3 V reference for the debug probe
2	IO_TX	IOMCU console transmit
3	NC	Not connected
4	SWDIO	IOMCU SWD data
5	SWCLK	IOMCU SWD clock
6	SWO	IOMCU trace output
7	GPIO1	IOMCU spare I/O
8	GPIO2	IOMCU spare I/O
9	nRST	IOMCU reset
10	GND	Ground

AF-H7E Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`. Mounting: Pixhawk FMUv6x Standard.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 4.75 – 5.7 V (Rated 5 V).
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-H7E Firmware

• Available Firmware

Type	Version	Date
ArduPilot Copter (.apj package)	1.3.0	2026-08-20
ArduPilot Copter - Bootloader + App (merged HEX / DFU / SWD)	1.3.0	2026-08-20
ArduPilot Plane (.apj package)	1.3.0	2026-08-20
ArduPilot Plane - Bootloader + App (merged HEX / DFU / SWD)	1.3.0	2026-08-20

Download files from the product page (QR).

AF-H7E Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Supply the servo rail (0 – 9.9 V) from an external BEC.
- Copter and Plane share board ID 6202 — pick the image by file name.



<Detailed Manual>

